Graphing Labrador — Full Website Report

Version described: the completed state after the final refinements round (22 October 2026).
Scope: the website as it actually exists in the project folder. Every statement below was checked against the current source code, the automated tests, or a recorded browser check. Where something is not built, it is listed as such.

Status labels used throughout

Label	Meaning
Fully implemented	Built, reachable in the website and covered by a test or a browser check
Partially implemented	Built in part, or built with a stated limitation
Planned, not built	Not available. Never described as working

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1. Project overview

Purpose. Graphing Labrador is a browser-based learning environment for Year 9 Mathematics. It teaches graphing through interactive graphs, explicit modelling, immediate feedback and a cosmetic reward system, and it gives teachers visible evidence of what students did and where they went wrong.

Target audience. Year 9 students, particularly those who are comfortable on a computer but disengaged by conventional teaching, and their mathematics teacher.

Mathematics focus. The only completed lab is Slope Lab, which teaches gradient and rates of change: reading coordinates, rise and run, calculating gradient, positive, negative and vertical lines, units and real-world rates, and explaining reasoning. The curriculum link is stored as a draft mapping to the NSW Mathematics K-10 (2022) Stage 5 syllabus. No official outcome codes appear anywhere, because they have not been validated by a teacher.

Educational philosophy. The design follows explicit teaching: the student sees a process modelled, practises it with support, then practises it independently. Rewards recognise effort and success but do not replace learning; everything needed to learn is free and the shop is cosmetic only.

Implementation status

Area	Status
Slope Lab lesson (8 stages, 11 questions)	Fully implemented
Student portal (7 pages plus name entry)	Fully implemented
Teacher portal (6 pages plus student profile)	Fully implemented for the demo data
Axis Lab, Vector Lab, Vertex Lab, Curve Lab lessons	Planned, not built
Production database, accounts and server-side scoring	Planned, not built (schema file only)

2. Design and identity

Name. Graphing Labrador is wordplay on graphing, Labrador (the breed) and lab (a place to experiment). The student-facing world is called Vector Lab.

Colour. A dark navy base, turquoise (brand colour) and amber/yellow (accent), with off-white text. Colours are CSS variables in `src/styles/index.css` for dark and light themes, so components use meaning-based names rather than fixed colours. The graph-paper pattern appears behind headers.

Mascots. Five original Labradors drawn as layered SVG code in `src/components/mascots/Labrador.tsx`. Each wears a collar badge showing a mathematical symbol.

Character	Nickname	Lab	Badge motif	Role
Vector	Hector	Vector Lab	Arrow	Main guide
Vertex	Tex	Vertex Lab	Parabola	Key graph features
Axis	Axel	Axis Lab	Axes	Coordinates and plotting

Character	Nickname	Lab	Badge motif	Role
Slope	Slobber	Slope Lab	Slope triangle	Gradient and rates
Curve	Kirby	Curve Lab	Wave	Non-linear relationships

The characters share one base drawing and differ in badge motif and accent colour. They have three expressions (happy, thinking, proud) and can wear hats, neckwear and eyewear, take a coat colour and sit on a background. Distinct poses and personalities beyond the badge are partially implemented: the personalities are described in data but only lightly visible in the artwork.

Themes and text. Dark and light themes and a larger-text toggle are in the sidebar (or the top bar on narrow screens). Animation is reduced automatically when the device asks for it.

Intended experience. The mascot is a presence, not a constant narrator: it appears on the dashboard, labs, kennel, shop, leaderboard and lesson header and never interrupts with pop-ups.

3. Technical architecture

Technologies (from `package.json`)

Purpose	Package
Interface	React 19, TypeScript 6
Build and dev server	Vite 8
Styling	Tailwind CSS 4
Routing	React Router 7 (hash routing)
Maths notation	KaTeX
Icons	lucide-react
Tests	Vitest 5, React Testing Library, jsdom
Lint	ESLint 10 with typescript-eslint

Recharts is still listed as a dependency but is no longer used by any page. It can be removed.

Structure

```
src/
  app/          App.tsx: the route table
  components/   assessments, graphs, layout, mascots, rewards, ui
  pages/        public, student, teacher
  hooks/        useStudent, useClassData
  lib/          math, scoring, rewards, analytics, lesson (no React inside)
  data/         curriculum, lessons, demo (seed data)
  services/     persistence (browser store and actions)
  styles/       index.css (design tokens)
  types/        shared types
tests/unit/     automated tests
supabase/       database schema and seed (not connected)
docs/           this report
```

How it works. Pages read state through `useAppState` and change it only through functions in `src/services/persistence/actions.ts`. The state is a single object kept in memory and mirrored to the browser's localStorage. Pure logic (graph maths, answer checking, reward rules, analytics) lives in `src/lib` and has no knowledge of React or storage, which is what makes it testable.

Routing. Hash routing (`#/student/...`, `#/teacher/...`) so the site works on static hosting without server rules. Student and teacher areas redirect to the landing page if the role is not signed in.

Demo versus production. Only demo mode exists. The production schema is a separate file and is not connected to the website.

4. Complete page directory

Public

Page	Route	Contents
Landing	`#/`	Title, description, Enter as Demo Student and Enter as Demo Teacher buttons, demonstration disclaimer, reset-demo-data button, the five Labradors, theme and text controls

Student portal

Page	Route	Contents and features
Name entry	shown inside `#/student` until a name is saved	"My name is..." screen. Requires a non-empty name (maximum 30 characters)
Dashboard	`#/student`	Welcome and Edit Name, level and XP bar, coins, daily check-in, Continue learning card, My Labs list, Active assignments, How rewards work
My Labs	`#/student/labs`	Five lab cards in the order Vector, Vertex, Axis, Slope, Curve. Only Slope Lab opens. The others show In Development

Page	Route	Contents and features
Lab page	`#/student/labs/:labId`	Lab description, mascot and lessons list
Slope Lab lesson	`#/student/lesson/gradient`	The 11-question lesson (section 6)
My Progress	`#/student/progress`	Level, XP, coins earned and spent, skill messages, released teacher feedback
My Kennel	`#/student/kennel`	Choose an unlocked character and equip owned cosmetics, reset appearance
Reward Shop	`#/student/shop`	64 cosmetics (All view sorted by price), filter by category, confirm purchase, insufficient-funds and owned states, purchase history
Leaderboard	`#/student/leaderboard`	Class ranked by XP using display names; hidden if the teacher turns it off

Teacher portal

Page	Route	Contents and features
Dashboard	`#/teacher`	Four summary cards, Accuracy by skill, Common errors, Missing work, Recent submissions, CSV export
Students	`#/teacher/students`	Class table with lesson status, questions attempted, last activity and flags
Student profile	`#/teacher/students/:id`	Diagnostic summary, skill evidence, misconception evidence with actions, interventions, notes, earlier-versus-later attempts, full attempt history
Lesson Library	`#/teacher/lessons`	Full Slope Lab teacher information, preview button, four collapsed In Development panels
Results and Marking	`#/teacher/marking`	Written responses to mark and already marked, with the rubric
Rewards Settings	`#/teacher/rewards`	Fixed question reward rules, editable daily check-in and lesson bonus, leaderboard switch

5. Student experience

1. **Entry.** The student chooses Enter as Demo Student. On first entry they see "My name is..." and must type a name. The student becomes the tenth member of the demonstration class alongside nine fictional classmates.
2. **Name.** The name shows in the sidebar, dashboard, progress page, leaderboard and every teacher view. Edit Name on the dashboard changes it without touching progress, coins, XP or results, because everything is linked by an internal student id.
3. **Dashboard.** Shows the next action (Start or Resume lesson), level and XP, coins, assignment status, labs and a plain-language reward guide. A daily check-in gives a small coin reward once per day.

4. **Lesson.** The lesson is a single vertical column at every screen width: instruction or worked example, then the graph, then the question, then the answer input, the submit button and the feedback. Nothing moves on without the student pressing **Next Question** (or Next step in the think-aloud).
5. **Resuming.** The lesson saves the stage, question and think-aloud position. Leaving and coming back returns to the same place with a "Welcome back" note. Earlier answers and hints are restored and rewards are not repeated.
6. **Completion.** A summary shows questions answered, coins and XP earned in the lesson, a next-step message and a Restart for revision button. A restarted lesson earns no repeat rewards.
7. **Spending.** Coins buy cosmetics in the shop. Items are equipped in My Kennel and appear on the dashboard, sidebar and labs.

6. Learning content

Labs

Lab	Character	Status
Slope Lab	Slobber	Fully implemented (one lesson)
Axis Lab	Axel	Planned, not built
Vector Lab	Hector	Planned, not built
Vertex Lab	Tex	Planned, not built
Curve Lab	Kirby	Planned, not built

Question types implemented and used

Type	How the student answers	Used in
Multiple choice	Choose one option	Questions 1, 6, 10
Numeric	Type a number, fraction or decimal, with units where required	Questions 2, 3, 4, 5, 7, 8
Plot a point	Click the graph, or type x and y	Question 9
Written explanation	Free text, marked by the teacher	Question 11

Two further types (read a point from a graph, and match items) exist in the code but are not used by any lesson. They are not part of the student experience.

The 11 Slope Lab questions

The lesson has eight stages. The think-aloud stage (I do) has no question. Total marks: 20.

No.	Stage	Question	Type	Marks	Purpose
1	Warm-up (hook)	What happens to the line when point B moves upward?	Multiple choice	1	Notice that moving a point up makes the line steeper. Student can drag point B first

No.	Stage	Question	Type	Marks	Purpose
2	We do	Going from P to Q, what is the rise? (answer -6)	Numeric	1	Rise as vertical change, including a negative rise
3	We do	What is the run from P to Q? (answer 3)	Numeric	1	Run as horizontal change
4	We do	Calculate the gradient, rise over run (answer -2)	Numeric	2	Combine rise and run; negative gradient
5	You do	Gradient of the line through R(-3,-1) and S(1,1) (answer 1/2)	Numeric	2	Independent calculation with negative coordinates and a fraction
6	Spot the mistake	Maya divided run by rise. What is her mistake?	Multiple choice	1	Recognise the reversal error in worked working
7	Real world	How quickly is the cyclist travelling? (answer 15 km/h)	Numeric with units	2	Gradient as a rate, with units
8	Checkpoint	Gradient through C(2,3) and D(5,12) (answer 3)	Numeric	2	Independent check with larger values
9	Checkpoint	Plot the point (-3, 2)	Plot a point	1	Coordinates with a negative x
10	Checkpoint	What is the gradient of a vertical line? (answer undefined)	Multiple choice	1	Zero versus undefined gradient
11	Reflection	What does gradient tell us that the height of a graph alone does not?	Written	6	Explain reasoning in words, marked with the rubric

The stage list shown to students is: Move the point, I do, We do, You do, Spot the mistake, Use it in the real world, Checkpoint, Reflect, then a Rewards summary.

7. Interactive graph engine

The graph is drawn as SVG by `src/components/graphs/CartesianGraph.tsx`. All mathematics is in `src/lib/math/graph.ts` and is independent of the drawing.

Feature	Detail
Coordinate mapping	Maths coordinates to screen pixels and back, with y pointing up; tested for round trips, boundaries, negatives and fractions
Axes	Positive and negative axes through the origin; labels and tick values; configurable range and step for each axis
Points	Static points, or draggable points that snap to the grid and stay within the plot area
Keyboard	A focused draggable point moves with the arrow keys; it announces its coordinates

Feature	Detail
Touch and mouse	Pointer events, so dragging works on touch screens
Controls	Zoom in and out, grid on and off, coordinate labels on and off, reset
Line and triangle	Draws the line through two points, clipped to the plot area. Optional dashed rise and run triangle
Gradient working	A typeset fraction showing the substitution, using the point labels. Vertical lines are reported as undefined. Horizontal lines give 0
Plotting	Click-to-plot with snapping, plus typed x and y fields as a keyboard alternative
Text alternative	An expandable text and table description of every graph

Answers are not revealed early. The gradient working and the rise and run labels stay hidden on the We do, You do, Spot the mistake, Real world and Checkpoint graphs until the question is answered. The warm-up and the I do worked example show them live, by design.

Mathematical notation. KaTeX renders fractions, subscripts, exponents and units (`src/components/ui/MathText.tsx`). Authors write `$...$` for inline and `$$...$$` for display maths. Rise is amber, run is teal and negative values are bracketed, bold and lightly highlighted, so a student can see why 1 - (-1) = 2. Colour is never the only cue. Every expression has a spoken-text alternative generated by `src/lib/math/tex.ts`.

8. Learning and teaching strategies

Strategy	Where it appears
Clear learning intention and success criteria	Lesson header (intention shown, criteria expandable)
Activating prior knowledge	Warm-up graph the student can move before any calculation
Explicit modelling / think-aloud	I do: six revealed steps stating the decisions before the calculation
Worked example	I do, with the A(1,2) and B(3,6) example and rise/run triangle
Gradual release (I do, We do, You do)	Stages in that order; no automatic hints in You do
Guided practice with feedback	We do: rise, run, then gradient on one line
Independent practice	You do and Checkpoint questions
Non-example and error analysis	Spot the mistake: a worked solution that contains an error
Application in context	Cyclist distance and time, with units
Formative checking	Immediate correct/incorrect feedback; unlimited retries
Student reasoning	Written reflection marked against a rubric
Optional scaffolds	Hints (recorded, never penalised); a hint about the student's own mistake on request

The teacher view of the lesson (Lesson Library) records the rationale: the worked example makes the process visible and reduces unnecessary working-memory demands; guided practice allows rehearsal with feedback; independent practice gives stronger evidence of unsupported understanding.

9. Literacy and numeracy

Literacy demands addressed

- Vocabulary: gradient, rise, run, rate, increase, decrease, positive, negative, undefined. Each question carries a recorded literacy demand.
- Reading questions precisely: for example, "how quickly" asks for a rate, not the highest point on the graph.
- Reading and writing coordinates in order, with signs.
- Explaining reasoning in words (the reflection).

Numeracy demands addressed

- Reading scales and negative coordinates.
- Subtracting signed numbers for rise and run.
- Dividing to give a gradient as a fraction or decimal.
- Interpreting a gradient as a rate with units from both axes.

How the website supports students

- Maths notation makes the structure of rise over run visible.
- The negative-value highlighting targets the common error in subtracting a negative.
- The text alternative and typed coordinate entry give a route that does not depend on dragging.
- Hints are staged: they start general and become more specific, and none appear automatically.

Limits. Support for students who need literacy or English-language help is limited to clear wording and hints. There is no glossary, no audio and no translated vocabulary (see section 18). Per-question literacy and numeracy demands are stored with every question but are shown only through the question bank, not on any current page.

10. Assessment and feedback

Automatic marking

Question type	How it is checked
Numeric	The number is parsed (integers, decimals, fractions, a minus sign, a leading plus) and compared with a tolerance of 0.01. Any trailing text is treated as units and checked separately
Numeric with units	Units are normalised so spelling variants match and different units do not
Multiple choice	The chosen option is compared with the correct option
Plot a point	The plotted point is compared with the target

Question type	How it is checked
Written	Never auto-marked. A valid response (at least 15 characters) is saved for the teacher

Equivalent answers accepted (expected 15 km/h): `15km/h`, `15 km/hr`, `15km/hr`, `15 kilometers per hour`, `15 kilometres per hour`, `15 kilometres/hour`, `15 km per hour`, plus differences in spaces, capital letters and decimal form (`15.0`) and equivalent fractions (`30/2`). **Not accepted:** the wrong value, the wrong sign (`−15`), a missing unit, or a wrong unit (`15 m/s`, `15 km`, `15 miles per hour`).

Feedback wording

Every automatically marked question shows exactly one of two messages and nothing else as the verdict:

- `Correct! Well Done!`
- `Incorrect! That's ok, give it another go!`

Rewards appear beside the message as separate coin and XP badges. A correct answer stays on screen until the student presses **Next Question**. An incorrect answer keeps the question open for another attempt, does not reveal the answer and does not show a hint automatically. After a wrong answer a student can ask for an optional hint about their mistake. After three wrong attempts, a student can ask for a worked example or skip the question for now. The worked example reveals the answer, but only when requested.

Error tagging

Wrong answers are tagged where the error is recognised: rise and run reversed (the reciprocal), a dropped negative sign, x and y reversed, a missing or wrong unit, zero given for a vertical line, and others. Tags feed teacher analytics and are never shown to the student as labels.

Written reflection and the rubric

The reflection is marked 0 to 2 on each of three criteria, for a total out of 6. Students can open **View Marking Rubric** beside the reflection box; teachers see the identical rubric with score buttons. It is one shared component, so the wording cannot differ.

The rubric text, word for word:

Example of a strong response
Height only tells you the value at one moment. Gradient tells you how quickly the value is changing, for example 15 km/h.

Correct mathematical interpretation
Strong: Correctly identifies graph features and explains their meaning in context.
Developing: Reads values correctly but does not explain what they represent.
Needs support: Misinterprets what gradient or other features mean.
Needs support (0) · Developing (1) · Strong (2)

Appropriate mathematical language and units
Strong: Uses correct vocabulary and units.
Developing: Uses some vocabulary; units inconsistent.
Needs support: Little mathematical language; units missing.
Needs support (0) · Developing (1) · Strong (2)

Explanation and justification
Strong: Explains why the answer is reasonable using evidence.
Developing: States a conclusion with limited support.
Needs support: No justification.
Needs support (0) · Developing (1) · Strong (2)

The "Example of a strong response" is a teacher-style example written for this project. It is not an official model answer.

Teacher marking. The teacher picks a level for each criterion, optionally writes feedback and presses Save and release. Saved marks stay after navigating or refreshing. Released feedback and the score out of 6 appear on the student's My Progress page. Marking is partially implemented in one respect: the rubric itself cannot be edited, and there is no assessment builder.

11. Rewards system

There are two separate values: **coins** (spendable in the shop) and **XP** (builds the level; every 100 XP is one level). Both are recorded in an append-only list of transactions. There is no third currency.

Per-question rules

Situation	Coins	XP
Correct on the first attempt	10	5
Incorrect on the first attempt	3	1
Incorrect on any later attempt	0	0
First correct answer after one or more incorrect attempts	5	2
Any attempt after the question has been completed	0	0

Worked totals: correct first time is 10 coins and 5 XP; wrong then correct is 8 coins and 3 XP; wrong, wrong, wrong, correct is also 8 coins and 3 XP.

Protection against duplicates. Rewards are checked against the student's saved attempts and the transaction list, keyed by question and student. Refreshing, navigating away, rapid clicking or replaying a completed question cannot pay again. This is covered by automated tests and was checked in a browser (a refresh after a correct answer left the coin total unchanged).

Other rules

- Invalid input (for example text where a number is needed) is not an attempt and earns nothing.
- A written reflection is never treated as correct. It earns the first-attempt amount (3 coins, 1 XP) once and nothing more.
- Daily check-in: 5 coins once per day (teacher-editable).
- Lesson complete: 40 coins and 50 XP once (teacher-editable).
- Starter balance: 30 coins, enough for the cheapest item.
- A wrong answer never removes coins or XP.
- The per-question amounts are fixed in code so the teacher settings page cannot contradict what students are told.

Shop

Sixty-four cosmetic items: 15 each of headwear, neckwear, eyewear and backgrounds, plus four characters. The shop never sells learning content. Coat colours were removed (any purchased earlier are refunded automatically, once, and unequipped).

Each of the four categories has the same spread of rarity: 5 common, 4 uncommon, 3 rare, 2 epic and 1 legendary. The legendary item in each category is animated (a floating, glowing halo; twinkling stardust collar; sparkling galaxy lenses; a slowly turning cosmos background). Animation stops when the device asks for reduced motion.

Category	Common (5)	Uncommon (4)	Rare (3)	Epic (2)	Legendary (1)
Headwear	Red Headband, Bucket Hat, Party Hat, Navy Beanie, Flat Cap	Beret, Lab Cap, Chef Hat, Cowboy Hat	Study Headphones, Wizard Hat, Pirate Hat	Gradient Crown, Viking Helmet	Golden Halo
Neckwear	Pink Ribbon, Grid Bandana, Jingle Bell, Blue Necktie, Bow Tie	Flower Collar, Teal Scarf, Striped Scarf, Star Medal	Flower Lei, Hero Cape, Pearl Necklace	Gold Chain, Amethyst Pendant	Stardust Collar
Eyewear	Eye Patch, Heart Glasses, Square Frames, Sports Visor, Round Glasses	Monocle, Aviators, 3D Glasses, Lab Goggles	Sunny Shades, Neon Frames, Steampunk Specs	Hologram Visor, Diamond Specs	Galaxy Lenses
Backgrounds	Graph Paper, Blue Sky, Mint Hills, Sandy Dunes, Plotted Points	Blueprint, Green Forest, Sunset Quadrants, Ocean Waves	Midnight Lab, Aurora, City Lights	Volcano, Deep Space	Living Cosmos

Category	Common (5)	Uncommon (4)	Rare (3)	Epic (2)	Legendary (1)
Characters			Axis (Axel), Slope (Slobber)	Vertex (Tex), Curve (Kirby)	

Prices run from 20 coins (cheapest background) to 900 coins (Living Cosmos). Legendaries cost 750 to 900 coins. The shop's All view lists every item from the lowest price to the highest. Purchases check ownership and balance, ask for confirmation and are recorded in a purchase history. A one-time item cannot be bought twice. Equipped items persist and show on the dashboard, sidebar and lab pages. Hats, glasses and neckwear are scaled to each dog's own head and neck.

Leaderboard

Ranks students by XP, using display names only. It does not use time spent. The teacher can switch it off. Real-world prize management is **planned, not built**.

Achievements

A separate Achievements page and badge system are **planned, not built**.

12. Teacher portal

The teacher is shown as Mr John Leary. The class is "Year 9 Maths 9A (demo)".

Dashboard. Four cards: Students, Checkpoint reached (percentage who have attempted every checkpoint question), Awaiting marking, and May need support (based on recorded evidence). Then:

- **Accuracy by skill.** Each skill's full name wraps naturally above its bar and percentage. It shows the share of auto-marked attempts that were correct across the class, including scaffolded attempts.
- **Common errors.** Eight graphing categories from the existing misconception system, each with a real count from recorded attempts. Categories with no recorded errors show a true 0.
- **Missing work.** Students who have not started, with a reminder to consider absence, access and support before escalating.
- **Recent submissions.** Short stage-level activity such as "Tomas completed 'We do'". It is built only from recorded attempts: a stage is "completed" only when every question in it has a correct (or submitted written) attempt, otherwise it reads "attempted". Written work shows to mark or marked. The think-aloud stage records no data so it never appears.
- **Export attempts (CSV)** with a guard against spreadsheet formula injection.

Students and student profile. The list shows each student's status, progress and flags. A profile shows a diagnostic summary, skill evidence, misconception evidence (with the suggested strategy, why it works, a follow-up task and how to check improvement), assign practice, mark for review, mark an intervention complete, teacher notes, first-versus-latest attempts and the full attempt history. Assign practice and mark for review create intervention records; there is no practice activity delivered to students yet.

Lesson Library. The complete Slope Lab teacher information: learning intention, success criteria, curriculum connection (draft), a typeset worked example, prerequisite knowledge, literacy and numeracy demands, teaching strategy, why it works, common misconception, suggested intervention, follow-up assessment, differentiation and

skills assessed. A preview button switches to the demo student. Below it, four collapsed panels (Axis, Vector, Vertex, Curve) carry the same field headings in the same order, each showing only a dash.

Results and Marking. Shown in section 10.

Rewards Settings. Shows the fixed question rewards; the daily check-in and lesson bonus are editable, with restore defaults; a leaderboard switch.

13. Learning analytics

What is recorded. For each attempt: student, question, lesson stage, skills assessed, the response, whether it was correct, marks, attempt number, hints used, an error tag where recognised, whether it awaits teacher marking, and the time.

Derived views

View	How it is produced
Skill level (no evidence, emerging, developing, secure)	From the latest result on each independent question. Independent means a You do, Real world or Checkpoint question answered with no hints. Scaffolded work (We do, warm-up, spot the mistake, or any use of hints) is reported separately
Provisional flag	Any conclusion based on fewer than three independent items is labelled provisional
Misconception evidence	One tagged error is "possible". The same error on two or more different questions is "likely". Each item links to the attempts behind it
Diagnostic summary	A plain sentence generated from the above
Needs-support flag	A likely misconception, or an emerging gradient-calculation skill

What a teacher can reasonably conclude. Which skills a student has answered correctly without support, which errors recur across questions, and where support was needed. Evidence is always one click from the underlying attempts.

Limits. The lesson has 11 questions, so most conclusions are provisional. Completion and time spent are not used as evidence of mastery. A wrong answer does not prove a misconception. The analytics make no claims about ability, learning needs or intelligence. The fictional classmates' data was generated by running the real scoring code on scripted behaviours; it illustrates the system and is not real student evidence.

14. Accessibility and differentiation

Implemented

- Skip-to-content link, visible focus outlines, labelled form fields and buttons, landmarks, and headings in order.
- Graph points can be moved with the arrow keys; coordinates can be typed instead of clicked; every graph has a text and table alternative.
- Maths expressions have spoken-text alternatives.
- Feedback is announced to screen readers (live regions).
- Status is not shown by colour alone (words and icons accompany it).

- Dark and light themes, a larger-text option, and reduced motion when the device requests it. No flashing effects.
- Charts on the teacher dashboard are text lists with bars, so labels are real text.

Differentiation. Optional staged hints that are recorded without penalty; a scaffolded We do before the independent You do; retry without limit; a revision restart; and a typed alternative to dragging. Scaffolds are optional and can be ignored by confident students.

Limitations (not yet done)

- Conformance to WCAG 2.2 AA is a design target. No formal audit and no testing with screen readers has been carried out.
- Dragging a point requires a pointer or the arrow keys.
- There is no glossary, audio, translation, or teacher-selectable adaptation.
- The written reflection has no spelling or speech support.
- Contrast was designed carefully but has not been measured with a tool.

15. Data storage and privacy

- **Where data lives.** In the browser's localStorage under one key (`graphing-labrador-demo-v1`). There is no server and no account system.
- **Not shared.** Nothing is synchronised between people, browsers or devices. The student and teacher roles are two views of the same browser's data.
- **Names.** The student types a preferred name and it is stored only in that browser. The nine classmates are fictional (the landing page says so).
- **No tracking.** There are no analytics, advertising or external services. Maths fonts are bundled with the site.
- **Reset.** The landing page has a Reset demo data button.
- **Upgrades.** Saved data is upgraded in place when the website changes, so existing progress is kept. Only the fictional classmates' data is regenerated.
- **Security caveat.** Because everything runs in the browser, a person using the site could edit their own saved data. Real security would need a server.

Production database (planned, not connected). `supabase/migrations/0001_initial_schema.sql` contains a database design with row-level security so that students could read only their own results and teachers only their own class. It has not been run, reviewed with a privacy team or connected. Server-side scoring has not been written.

16. Responsive design

- **Method.** Tailwind CSS with breakpoints at 640, 768 and 1024 pixels. Below 768 pixels the left sidebar becomes a top bar with a scrollable menu.
- **Lesson.** One column at every width, in the order: instruction, graph, question, input, submit, feedback.
- **My Labs.** An automatic grid whose cards never overlap; it reflows from one column to several.
- **Charts.** Text rows that wrap, so nothing is clipped.
- **Checked.** In a browser, at widths of 360, 768, 1024 and 1440 pixels, every student page and every teacher page had no horizontal scrolling. The lesson graph appeared above the question at each of those widths. The lab cards were also checked for overlap at seven widths between 320 and 1600 pixels during an earlier round.

- **Known limits.** Not tested on physical phones or tablets or on browsers other than the one used for checking. The graph is wide and is scaled down on narrow screens, which makes small points harder to grab.

17. Testing and quality assurance

Automated checks run for this report

Check	Result
TypeScript type check	Pass
ESLint	Pass
Production build	Pass
Unit and component tests	214 tests in 7 files, all pass

What the tests cover: graph maths (coordinates, gradient, intercepts, line clipping); answer checking and units, including all the equivalent forms listed in section 10; the exact feedback messages for every auto-marked question; every reward scenario and replay safety; saved-data upgrades; name entry and renaming without losing progress; lesson feedback not advancing by itself; the graph being above the question; lesson stages and question count; rubric text word for word and identical for students and teachers; the demonstration disclaimer; lab cards and wording; the teacher name; the Lesson Library panels; the dashboard activity and error calculations; and a check that no known misspelling appears anywhere in the source.

Browser checks performed (using browser automation against the running site)

- A fresh student answered a question correctly: the message and reward badges stayed on screen for at least 1.5 seconds with a Next Question button, and a page refresh kept the same question and the same coin total.
- A wrong, wrong, correct sequence on a numeric question paid 3 coins and 1 XP, then nothing, then 5 coins and 2 XP, and did not advance until Next Question was pressed.
- No horizontal scrolling on any student or teacher page at four widths.
- The teacher dashboard showed full skill names, eight error categories with zeros and short stage-level activity.

Not tested

- There is no Playwright test suite. The checks above were run by hand through an automation tool and are not repeatable by script.
- Real devices, other browsers, screen readers and an accessibility audit.
- Performance, and long-term storage limits.
- The production database and schema.

18. Incomplete features

Planned, not built

- Axis Lab, Vector Lab, Vertex Lab and Curve Lab lessons.
- Accounts, a database connection and server-side scoring.
- Assessment builder, assignment creation, due-date changes, reminders and parent-contact preview.
- Rubric editor.

- Achievements page and badges; Graph Studio; Missions page; a Settings page.
- Printable teacher reports and filters (only the CSV export exists).
- Real-world prize approvals and teacher-awarded unlocks.
- Glossary, audio, translations and teacher-selectable adaptations.
- Automated browser test suite.
- Validation of syllabus mappings and official outcome codes.

Partially implemented

- Mascot personalities (badge and colour only).
- Teacher interventions (records are created; nothing is delivered to students).
- Accessibility (built in, not audited).
- Question types: matching and read-a-point exist in code but are unused.

19. Known issues and risks

Issue	Detail
Browser-only data	Clearing site data removes progress. Data is not backed up or shared
Not secure	A user can edit their own saved data; there is no authentication
Provisional analytics	Eleven questions give little evidence; conclusions are marked provisional
Worked example reveals the answer	Available after three wrong attempts on request
Unused code	Matching and read-a-point question support, a small line drawing component, and the Recharts dependency are unused
Demo due date	The assignment due date is set relative to first use and stored in the browser
Seeded classmates	Regenerated when the data version changes; their teacher marks are kept only for responses that still exist
Automation limitation	Browser-automation clicks can time out when the page is not visible; the checks above worked around this
Spelling	The landing-page disclaimer deliberately contains "fitcional" (one isolated sentence, kept as requested). All other text was spell-checked and corrected (section 23)
Not production-ready	Needs school approval, privacy review, accounts and server-side validation before real students use it

20. Deployment and maintenance

Run locally (Node.js 20 or later; the project was built with Node 24)

```
 npm install
npm run dev        # http://127.0.0.1:5173/
npm run build      # type check, then production build into dist/
npm run preview    # serve the built site on 127.0.0.1:4173
npm run lint
npm run typecheck
npm test
```

Hosting. The build is static files in `dist/` and uses hash routing, so it can be served from any static host. For a sub-path (for example GitHub Pages) build with `VITE_BASE=/name/`.

Maintenance notes

- Lesson content is data in `src/data/lessons/gradient.ts`. Maths is written between dollar signs.
- Reward rules are one object in `src/lib/rewards/economy.ts` and one description component, `src/components/rewards/RewardRules.tsx`.
- If the question bank or seeded classmates change, increase `SEED_VERSION` in `src/data/demo/seed.ts` so saved browsers refresh the classmates only.
- A local Git checkpoint was created before this round of changes.

Required before real classroom use

1. School approval and a privacy impact assessment; a student data-handling agreement.
2. Real accounts, a hosted database, and scoring and rewards enforced on the server.
3. A child-safety review, data retention rules and an account removal process.
4. Teacher validation of the curriculum mapping; an accessibility audit with assistive technology.
5. Rules for any real-world prizes, set by the school.

21. Source-code reference

Feature	Main files
Routes and layout	`src/app/App.tsx`, `src/components/layout/Layouts.tsx`
Landing page and disclaimer	`src/pages/public/Landing.tsx`, `src/data/copy.ts`
Name entry and editing	`src/pages/student/NamePrompt.tsx`, `src/pages/student/Dashboard.tsx`, `setStudentName` in `src/services/persistence/actions.ts`
Lesson page, resume and revision	`src/pages/student/GradientLessonPage.tsx`, `src/lib/lesson/position.ts`
Lesson content	`src/data/lessons/gradient.ts`
Question card, feedback, hints	`src/components/assessments/QuestionCard.tsx`, `src/lib/scoring/messages.ts`
Answer checking and units	`src/lib/scoring/evaluate.ts`, `src/lib/scoring/parse.ts`

Feature	Main files
Graph drawing and controls	`src/components/graphs/CartesianGraph.tsx`
Graph mathematics	`src/lib/math/graph.ts`
Maths notation	`src/components/ui/MathText.tsx`, `src/lib/math/tex.ts`, `src/styles/index.css`
Reward rules	`src/lib/rewards/economy.ts`, `src/components/rewards/RewardRules.tsx`
Shop and kennel	`src/data/curriculum/shop.ts`, `src/pages/student/Shop.tsx`, `src/pages/student/Kennel.tsx`
Mascots	`src/components/mascots/Labrador.tsx`, `src/data/curriculum/labs.ts`
My Labs and lab pages	`src/pages/student/LabsMap.tsx`, `src/pages/student/LabDetail.tsx`
Rubric	`src/components/assessments/RubricView.tsx`, `src/pages/teacher/Marking.tsx`, `RUBRIC` in `src/data/lessons/gradient.ts`
Teacher dashboard	`src/pages/teacher/Dashboard.tsx`, `src/components/ui/BarList.tsx`, `src/lib/analytics/activity.ts`, `src/lib/analytics/errors.ts`
Student profile and analytics	`src/pages/teacher/Students.tsx`, `src/lib/analytics/diagnostics.ts`, `src/hooks/useClassData.ts`
Lesson Library and In Development panels	`src/pages/teacher/LessonLibrary.tsx`, `src/data/curriculum/inDevelopment.ts`
Skills and misconceptions	`src/data/curriculum/skills.ts`
Browser storage and upgrades	`src/services/persistence/store.ts`, `src/services/persistence/actions.ts`
Demo classmates	`src/data/demo/seed.ts`
Database design (not connected)	`supabase/migrations/0001_initial_schema.sql`
Tests	`tests/unit/`

22. Final change log

Changes made in this round:

1. **Feedback no longer skips ahead.** The lesson's saved position is now fixed when the page opens. Before, a lesson with no saved position recalculated its place after each answer and jumped to the next question on its own. The button is now labelled **Next Question**, and coins and XP appear as two separate badges. The two feedback messages are unchanged. Reward rules are unchanged and were re-verified.

2. **Vertical lesson layout.** The lesson is one centred column at all widths: instruction or worked example, graph, question and answer, then feedback. The graph's operation and the mathematics are unchanged.
3. **Accuracy by skill.** Replaced the clipped chart with wrapping text rows, so full skill names show at every width.
4. **Common errors.** Now shows eight graphing categories (rise and run reversed, incorrect negative gradient, x/y reversal, misreading graph scales, confusing intercept and gradient, confusing height with rate, failing to interpret units, comparing unequal scales), counted from real attempts. Three have recorded errors in the demo data; categories with none show 0. Added tag mappings for wrong units, intercept confusion and scale misreading.
5. **Recent submissions.** Now shows student and lesson stage ("completed", "attempted" or "submitted", with to mark or marked for written work), built from recorded attempts only.
6. **Lesson Library.** The four In Development labs are now compact collapsed panels with the same 14 field headings as Slope Lab in the same order, each showing a dash. The Slope Lab information is unchanged.
7. **Developer references removed.** A visible mention of the progress file in Rewards Settings was removed. The My Progress page and progress tracking are untouched. No other internal references were found in the interface.
8. **Six spelling slips added (later corrected).** Six deliberate slips were placed in short descriptive text in this round. In the next round they were all corrected (see section 23), because each sat in a reusable interface description.
9. **Tests.** Added tests for non-advancing feedback, the four reward scenarios, layout order, activity and error calculations, Lesson Library panels and text hygiene. Total 214 passing.
10. **Documentation.** Added this report and a Git checkpoint taken before the changes.

Not changed: the 11 questions and 8 lesson stages, the reward amounts, the marking rubric text, the mathematics rendering and the saved student data.

23. Later changes: mascot redesign and spelling audit

Mascots. The five Labradors were redrawn as five separate designs in `src/components/mascots/dogs.tsx`, composed by `src/components/mascots/Labrador.tsx` (same props as before, so every page that shows a mascot updates automatically). Hats, glasses and neckwear from the shop are mapped onto each dog's own head size.

Character	Design
Hector	Large, broad head, wide muzzle, relaxed ears, confident brows, chocolate coat, broad shoulders
Tex	Small, narrow face, oversized floppy ears, tongue out, large playful eyes, pale-yellow coat, small body
Axel	Lean, long muzzle, alert almond eyes, asymmetrical ears (one folded, one lifted), black coat with a light rim so it shows on the navy background
Slobber	Round face, floppy ears, wide cheerful smile, golden coat, stocky body
Kirby	Soft round face, short muzzle, gentle eyes, long ears set high with a flipped tip, cream coat, compact body

Each keeps its name, accent colour and mathematical collar badge. The shop coat-colour items (Chocolate, Black) recolour a dog but keep its own shape.

Spelling audit. Every user-visible string in `src` was checked against a dictionary and then reviewed by hand. Australian spellings (colour, practise, organised, modelled, labelled, travelling, penalised, kilometre) are correct and were kept.

Misspelling	Location	Reusable?	Action
seperate	My Labs header	Generic page description	Corrected to "separate"
automaticaly	My Kennel subtitle	Generic page description	Corrected to "automatically"
dashbord	Reward Shop description	Generic page description	Corrected to "dashboard"
sucesses	Leaderboard subtitle	Generic page description	Corrected to "successes"
comparisions	My Progress subtitle (shown with the student's name)	Template	Corrected to "comparisons"
absense	Teacher dashboard, Missing work note	Generic note	Corrected to "absence"
fitcional	Landing-page demonstration disclaimer	No: one isolated sentence written by the project owner	Retained as requested earlier. It can be changed in `src/data/copy.ts`

No other misspellings were found in the interface, lesson data, question bank, feedback messages, rubric, marking pages, rewards text, navigation or the seeded classmates' responses.

24. Later change: shop expansion

Coat colours were removed. Headwear, neckwear, eyewear and backgrounds now each have 15 items with a 5 / 4 / 3 / 2 / 1 split across common, uncommon, rare, epic and legendary, and each category's legendary is animated. The All view is sorted by price. Artwork is in `src/components/mascots/cosmetics.tsx`, the catalogue in `src/data/curriculum/shop.ts`, animation in `src/styles/index.css`, and the saved-data upgrade (refund and unequip of removed coat colours) in `src/services/persistence/store.ts`. Tests are in `tests/unit/shop.test.tsx`.